

LAB ASSIGNMENT - 01

Experiment Title:-

Study of LAN Topology. Network Devices and Address configuration

Questions:-

1) Draw the Network Topology:

- Identify and draw the LAN topology of your organization
- Clearly label all components such as router, switch, access points and end devices (Pcs, laptops etc)

Solution:-

The Academic campus network uses a hybrid hierarchical topology that combines three distinct structural topologies. It combines:

- Tree Topology
- Star topology
- Ring topology.

This network is further divided into three-layered network model which are

- core layer
- Distribution layer
- Access layer

i) Core layer

• Location : Server room (Room 210, 2nd floor)

• Equipments :

i) Router and Firewall:-

Manages external ISP connections, security filtering and external gateway routing.

i) Layer - 3 (3+) switches:-

These switches are used in ring loop.

ii) Distribution layer :-

The main function is to interconnect the access floor layer and enforces local routing.

It consists of 4 distribution switches

- i) R1 (Room 213. 2nd floor): Direct link to server room.
- ii) R2 (Room 201. 3rd floor): aggregates upper floor traffic
- iii) R2 (Room 201. 2nd floor): Interconnected with R1 and R7
- iv) R6 (Room 113. 1st floor): aggregates lower floor and ground floor units.

Components used:-

- 9P-Link R-SG2 218 MPE
- 9P-Link 8 made 8034 28 x MP.

iii) Access layer:-

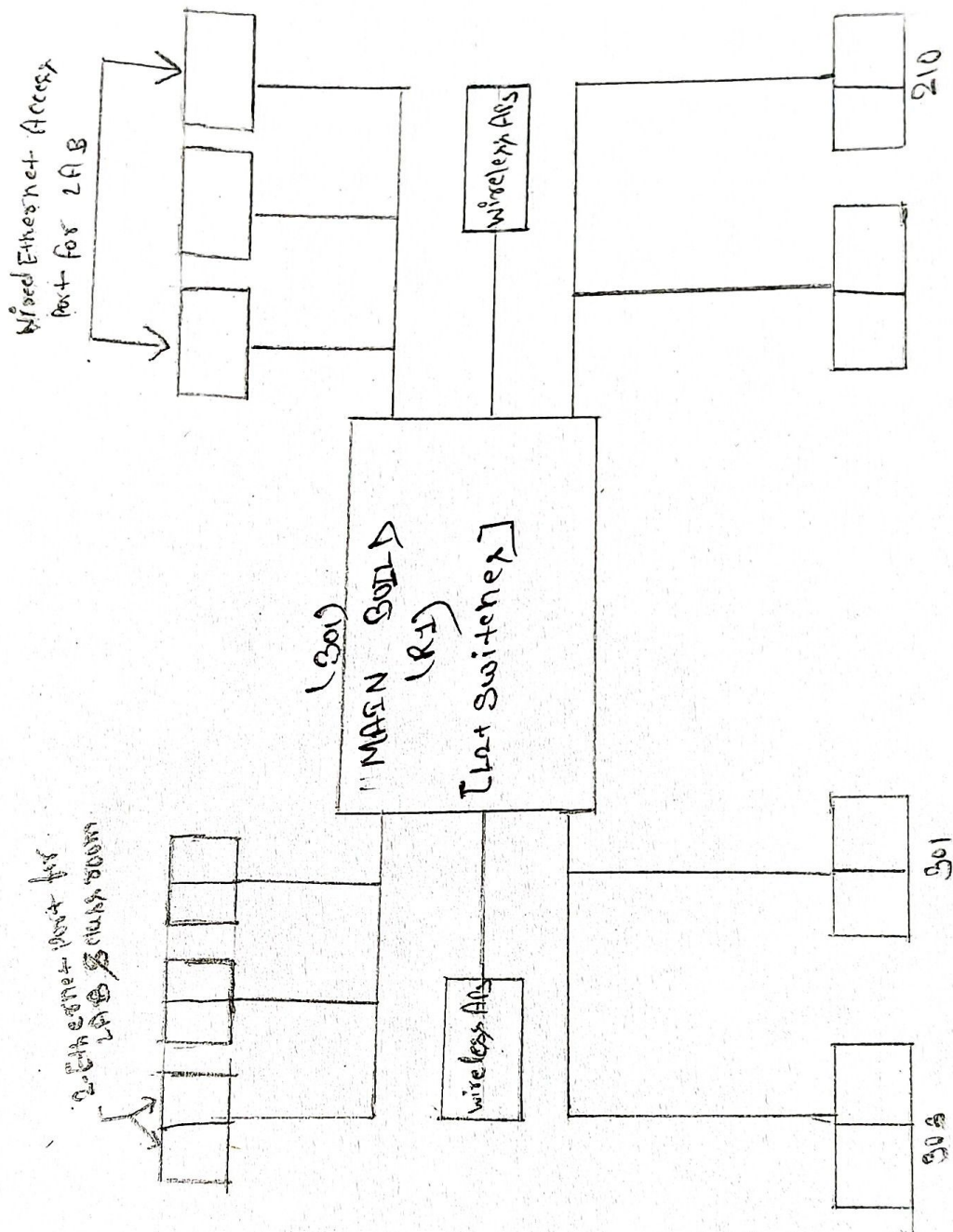
The access layer directly connects end-user devices connected devices

- classroom Ethernet ports
- Laboratory systems.
- Faculty Room PCs
- Wireless Access Points (APs)
- Outdoor wireless APs.

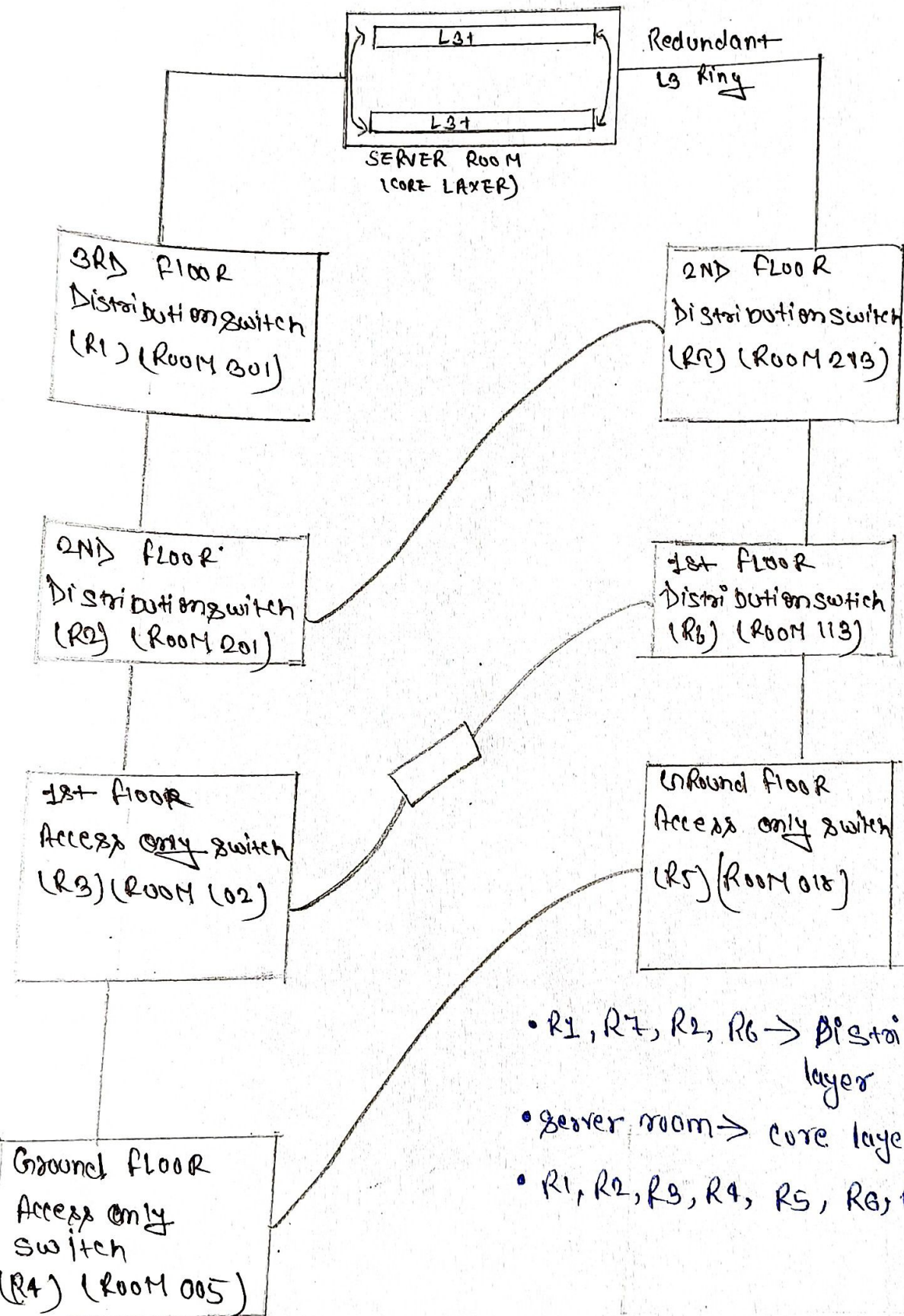
Access layer switches:-

- R1 → 3rd floor
- R2 → 2nd floor
- R3 → 1st floor
- R4 → Ground floor
- R5 → Ground floor
- R6 → 1st floor
- R7 → 2nd floor

Mixed Ethernet Access
Port for 2A B



ACADEMIC NETWORK TOPOLOGY



Question-02:- Identify Network Devices

- List all networking devices used in the LAN
- Mention the function of each device.

Solution:-

1.) Router and Firewall:-

Location:- Central Server room (Room 210, 2nd floor)

Function

- Acts as the primary gateway connecting the internal campus LAN to the external Internet service providers
 - It directs incoming and outgoing data traffic using IP routing.
 - It filters data packets to protect the school network from unauthorized external threats.
- 2.) Layer 3 Core Switches:-

Location:- Distributed in central server room it is deployed as a dual-switch redundant pair

Functions:-

- It forms the main high-speed backbone of the entire campus network.
- Handles faster inter-VLAN routing (routing data between different floor, labs and departments)
- Handles faster inter-switches. These switches are configured in ring topology loop to provide physical link redundancy.

3. TP-Link Omada S3428XMP (24x24 2+8 switches):-
Location:- Distribution layer nodes : R1, R2, R6, R7
Function:-

- (i) Aggregates traffic from the ~~lga~~ local access layer switches and end user devices on each floor before uploading it to the core layer.
- (ii) Provides managed layer 2 capabilities like traffic prioritizing (QoS), security policies and network division.
- (iii) It supports inline PoE+ (Power over Ethernet) power directly to downline access points through its 24 Gigabit ports.

4. TP-Link TL-S31218MPE:-

Location:- Access layer nodes : R3, R4, R5
Function:-

- (i) Connects edge devices directly into the school network infrastructure.
- (ii) It offers 16 dedicated PoE+ ports specifically optimized to run network periphery that require both data and power simultaneously.

5. Wireless Access Points (APs):-

Location:- ~~Access layer nodes : R3, R4, R5~~
Distributed access all floors and out door zones.
Function:-

- (i) Connects edge devices directly into the school network.
- (ii) It emits radio signals to provide wireless local area network (WLAN) coverage across classrooms, labs etc.
- (iii) Bridges mobile wireless client devices (laptops, smart phones, tablets) back into the physical wired ethernet distribution layer.

QUESTION 3: Analyze Connection Types:

- Specify whether connections are:
 - Wired(Ethernet)
 - Uses cable connection
 - Used in offices, labs, servers
 - Wireless (Wi-Fi)
 - Uses wireless signals
 - Used in mobile,laptop,home internet
- Mention where each type is used.

```
ritu@ritu-HP-Laptop-15s-fr4xxx:~$ ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 916 bytes 80095 (80.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 916 bytes 80095 (80.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlo1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.201.241.179 netmask 255.255.255.0 broadcast 10.201.241.255
    inet6 fe80::ff57:ccfc:bbdd:300b prefixlen 64 scopeid 0x20<link>
    inet6 2409:40e4:2e:d784:b418:bce5:1212:9cdf prefixlen 64 scopeid 0x0<global>
    inet6 2409:40e4:2e:d784:6dca:3a67:9160:b876 prefixlen 64 scopeid 0x0<global>
    ether 74:97:79:d1:49:15 txqueuelen 1000 (Ethernet)
    RX packets 4347 bytes 4900479 (4.9 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1633 bytes 372657 (372.6 KB)
    TX errors 0 dropped 6 overruns 0 carrier 0 collisions 0

ritu@ritu-HP-Laptop-15s-fr4xxx:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 74:97:79:d1:49:15 brd ff:ff:ff:ff:ff:ff
    altname wlp1s0
    altnam wlx749779d14915
    inet 10.201.241.179/24 brd 10.201.241.255 scope global dynamic noprefixroute wlo1
        valid_lft 3315sec preferred_lft 3315sec
    inet6 2409:40e4:2e:d784:b418:bce5:1212:9cdf/64 scope global temporary dynamic
        valid_lft 6987sec preferred_lft 6987sec
    inet6 2409:40e4:2e:d784:6dca:3a67:9160:b876/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 6987sec preferred_lft 6987sec
    inet6 fe80::ff57:ccfc:bbdd:300b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ritu@ritu-HP-Laptop-15s-fr4xxx:~$ iwconfig
Command 'iwconfig' not found, did you mean:
  command 'iconfig' from deb ipmiutil (3.2.1-1)
  command 'ifconfig' from deb net-tools (2.10-1.3ubuntu2)
Try: sudo apt install <deb name>
ritu@ritu-HP-Laptop-15s-fr4xxx:~$ nmcli dev wifi
IN-USE  BSSID          SSID          MODE  CHAN  RATE        SIGNAL  BARS  SECURITY
*       86:52:3B:8D:9C:CF  realme 15 SG p3q7  Infra  4      117 Mbit/s  84      █ WPA2 WPA3
ritu@ritu-HP-Laptop-15s-fr4xxx:~$
```

->By using ipconfig

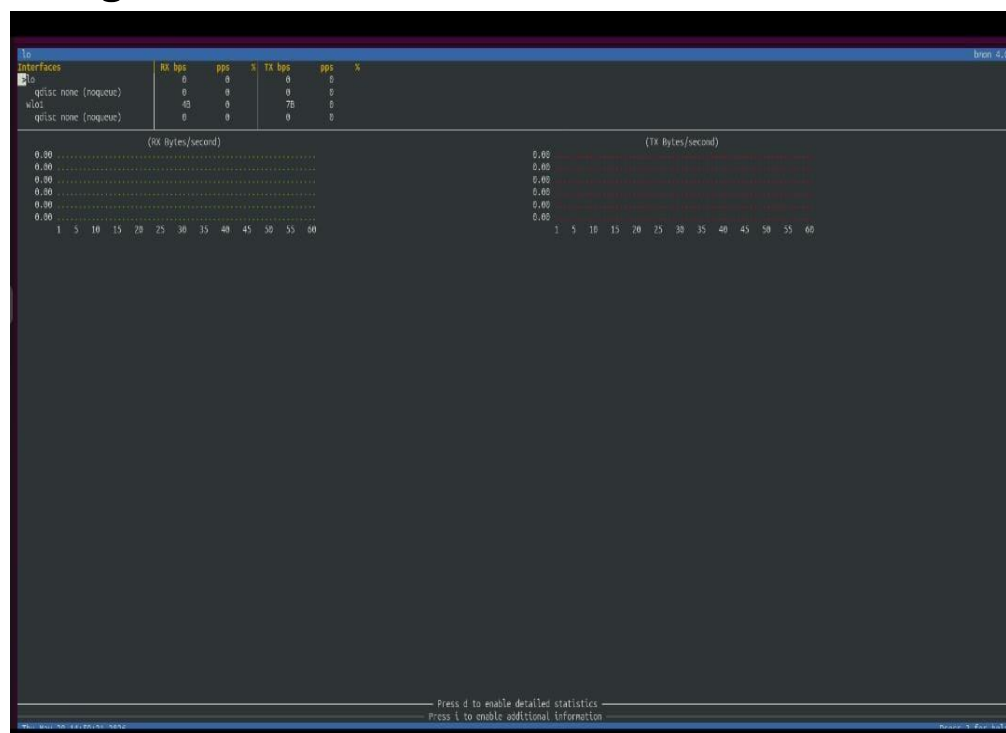
QUESTION 4:- Measure Network Speed:

Wireless Network

- Wireless interface: wlo1
- Link speed: 117 Mbit/s
- Network traffic monitored using bmon

Observation

- Wi-Fi connection was active.
- Live upload/download traffic was displayed using bmon.



->By using nmcli dev wifi
bmon

QUESTION 5:- Find System Network Details:

- Determine and record the following of your computer:
- Commands Used
 - > MAC Address → ip link
 - > IP Address → ip addr
 - > Subnet Mask → ifconfig or ip addr

```
ritu@ritu-HP-Laptop-15s-fr4xxx:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 74:97:79:d1:49:15 brd ff:ff:ff:ff:ff:ff
    altname wlp1s0
    altname wlx749779d14915
    inet 10.201.241.179/24 brd 10.201.241.255 scope global dynamic noprefixroute wlo1
        valid_lft 2272sec preferred_lft 2272sec
    inet6 2409:40e4:2e:d704:b418:bce5:1212:9cdf/64 scope global temporary dynamic
        valid_lft 6892sec preferred_lft 6892sec
    inet6 2409:40e4:2e:d704:6dca:3a67:9168:b876/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 6892sec preferred_lft 6892sec
    inet6 fe80::ff57:ccfc:bdd:3006/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ritu@ritu-HP-Laptop-15s-fr4xxx:~$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DORMANT group default qlen 1000
    link/ether 74:97:79:d1:49:15 brd ff:ff:ff:ff:ff:ff
    altname wlp1s0
    altname wlx749779d14915
ritu@ritu-HP-Laptop-15s-fr4xxx:~$
```

```

ritu@ritu-HP-Laptop-15s-fr4xxx:~$ ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1272 bytes 123184 (123.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1272 bytes 123184 (123.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlo1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.201.241.179 netmask 255.255.255.0 broadcast 10.201.241.255
    inet6 fe80::ff57:ccfc:bbdd:300b prefixlen 64 scopeid 0x20<link>
    inet6 2409:40e4:2e:d784:b418:bce5:1212:9cdf prefixlen 64 scopeid 0x0<global>
    inet6 2409:40e4:2e:d784:6dca:3a67:9160:b876 prefixlen 64 scopeid 0x0<global>
    ether 74:97:79:d1:49:15 txqueuelen 1000 (Ethernet)
    RX packets 39277 bytes 44745336 (44.7 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12917 bytes 4702979 (4.7 MB)
    TX errors 0 dropped 6 overruns 0 carrier 0 collisions 0

ritu@ritu-HP-Laptop-15s-fr4xxx:~$ nmcli device wifi
IN-USE BSSID SSID MODE CHAN RATE SIGNAL BARS SECURITY
* 86:52:38:8D:9C:CF realme 15 5G p3q7 Infra 4 117 Mbit/s 88  WPA2 WPA3
ritu@ritu-HP-Laptop-15s-fr4xxx:~$ iw dev
phy#0
    Interface wlo1
        ifindex 2
        wdev 0x1
        addr 74:97:79:d1:49:15
        type managed
        channel 4 (2427 MHz), width: 20 MHz, center1: 2427 MHz
        txpower 20.00 dBm
        multicast TXQ:
            qsz-byt qsz-pkt flows drops marks overlmt hashcol tx-bytes tx-packets
            0 0 0 0 0 0 0 0 0
ritu@ritu-HP-Laptop-15s-fr4xxx:~$

```

->By using iwconfig

->Nmcli device wifi

->Iw dev

Question - 06 Explain observations:

Briefly Explain:-

→ Difference between wired & wireless speed.

	Wired Speed	Wireless Speed
Data Flow	Wired supports full-duplex transmission (simultaneous two-way of transmission)	Wireless on the other hand operates on half duplex (only one way at a time)
Signal Interference	Zero interference from physical obstacles or other radio devices.	High interference from walls, microwave ovens and neighboring routers.
Signal Transmission	Electrical or light pulses inside a shielded cable.	Radio frequencies (2.4 GHz, 5 GHz, 6 GHz) through open air
Latency	Wired paths have minimal delay.	Wireless suffers from signal overhead and packet loss.

→ Explain role of Subnet Mask in networking.

Solution:-

A subnet mask is a 32-bit number used to manage network traffic routing. It identifies the network host portion of an IP address.

• Separation:- It splits IP address into two parts.

• Host ID:- It identifies the specific device address.

• Network ID:- It identifies the specific network address.

• Routing:- It tells router if a destination is local or it is external.

• Efficiency:- It prevents unnecessary broadcast traffic across networks.